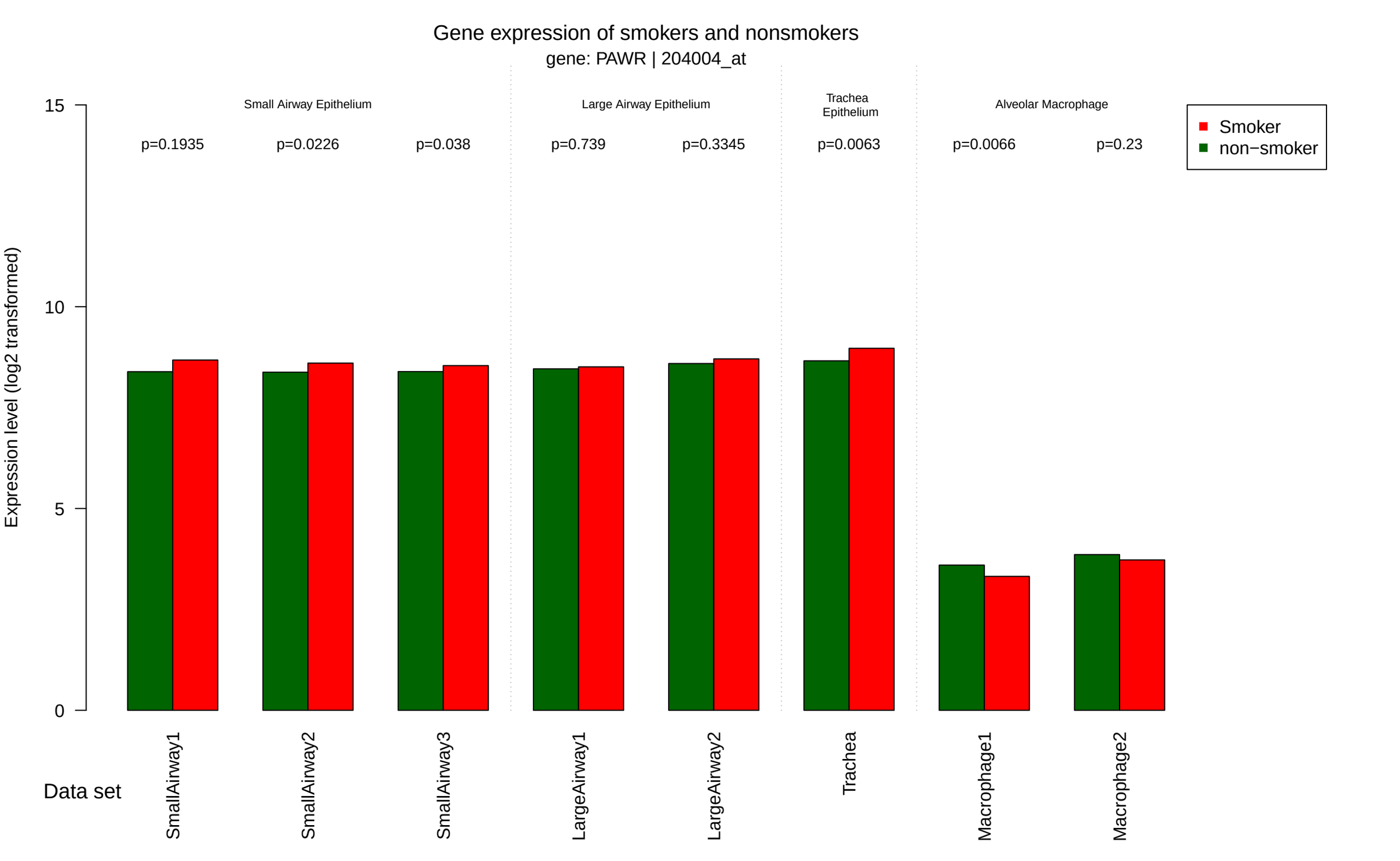
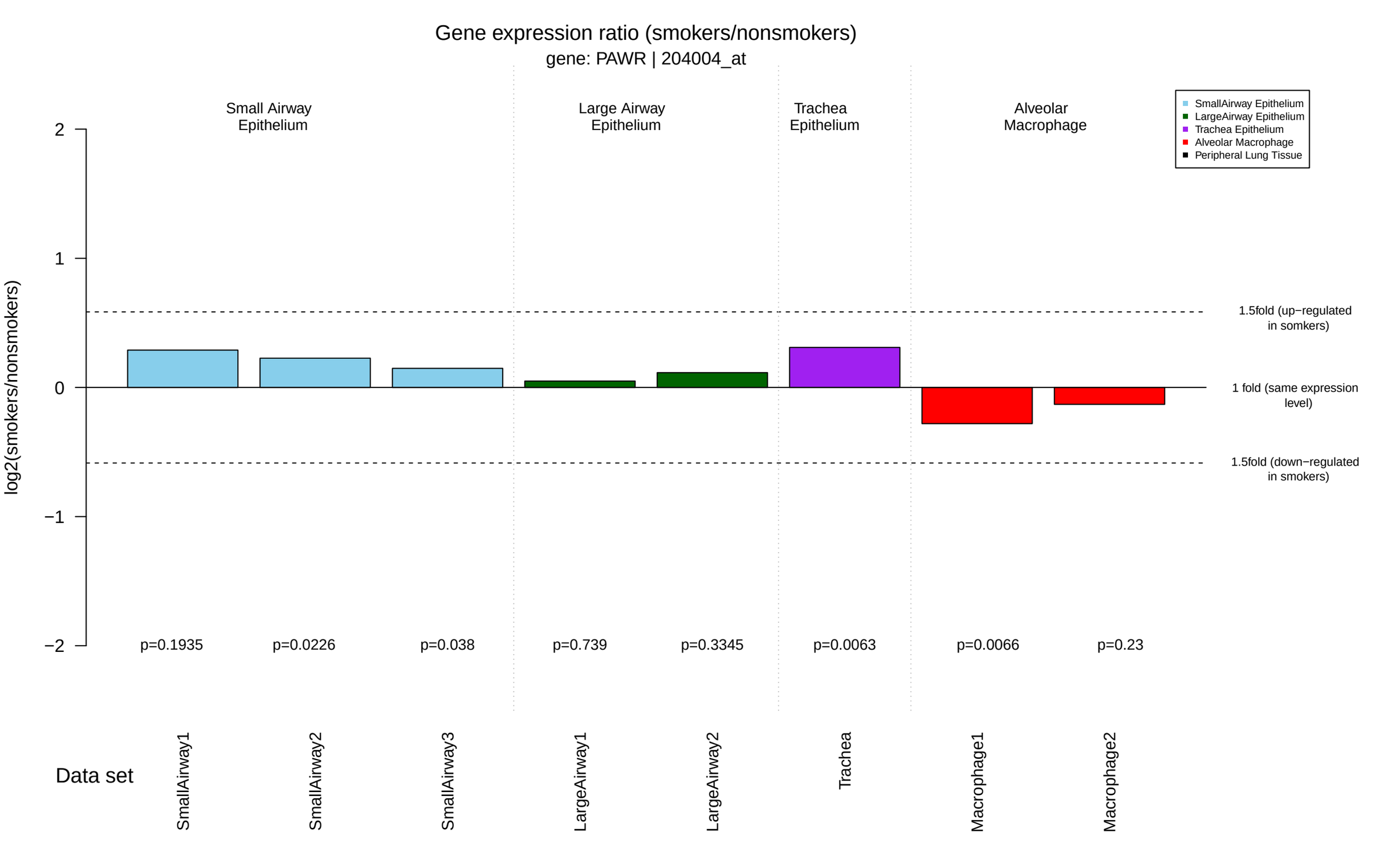
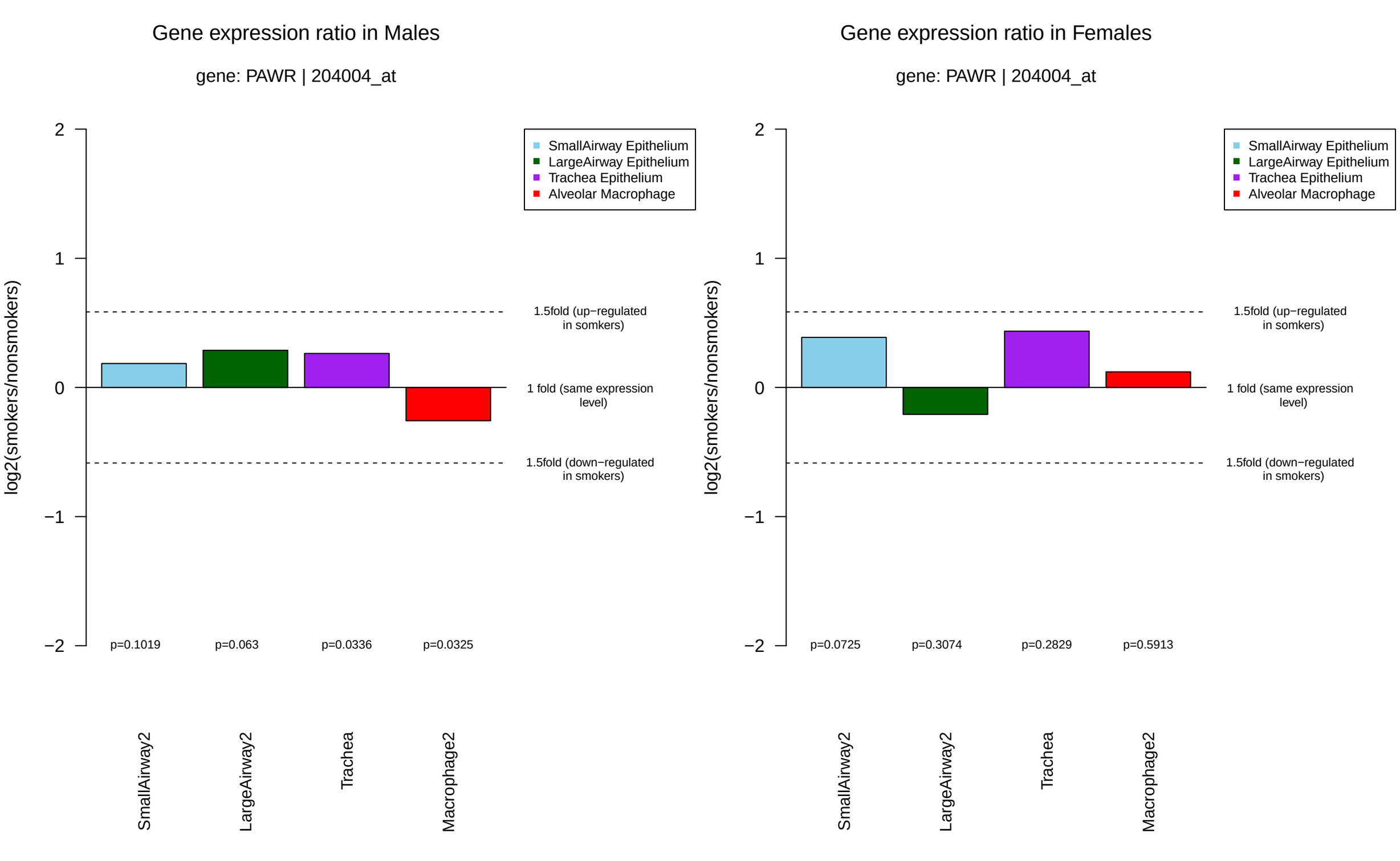
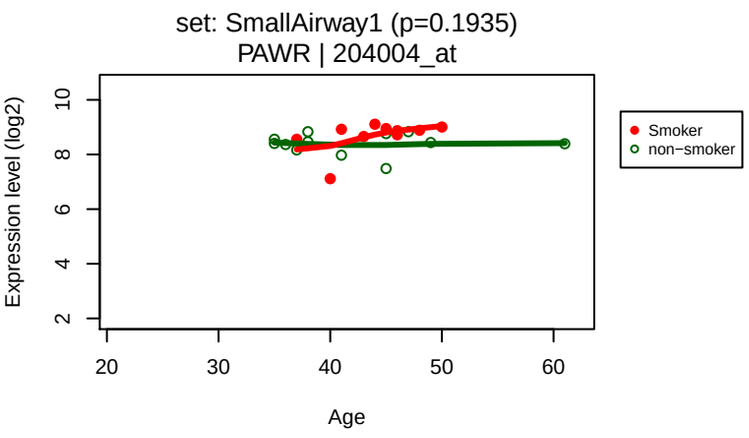
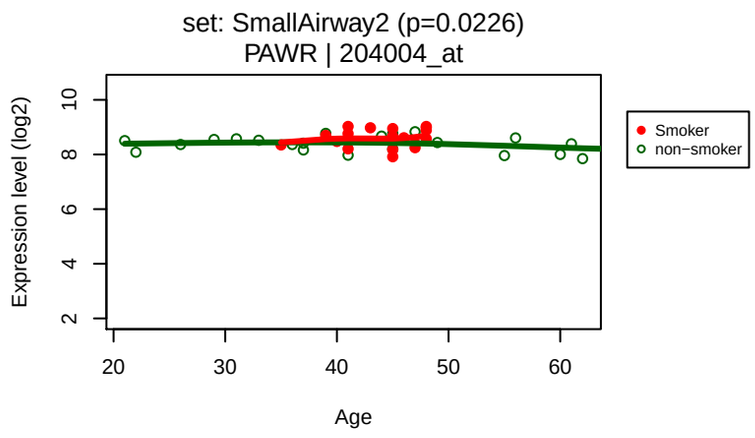
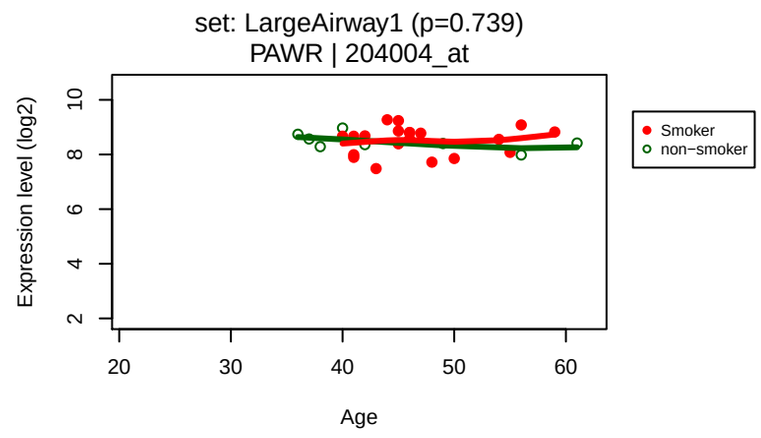

Gene expression of smokers and nonsmokers
gene: PAWR | 204004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1935
p=0.0226
p=0.038
p=0.739
p=0.3345
p=0.0063
p=0.0066
p=0.23
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

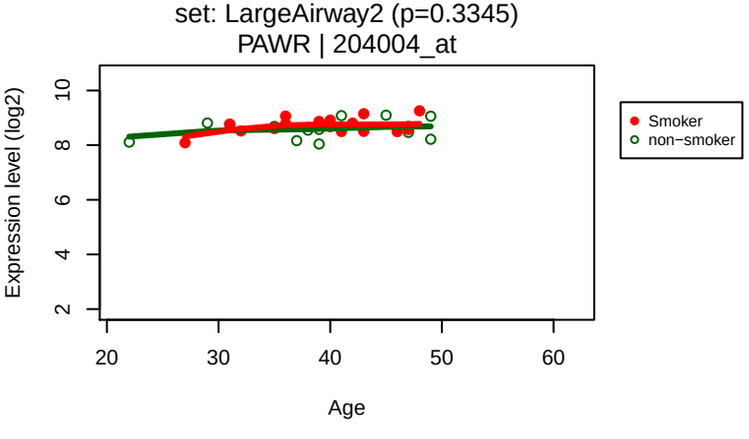

Gene expression ratio (smokers/nonsmokers)
gene: PAWR | 204004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1935
p=0.0226
p=0.038
p=0.739
p=0.3345
p=0.0063
p=0.0066
p=0.23
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

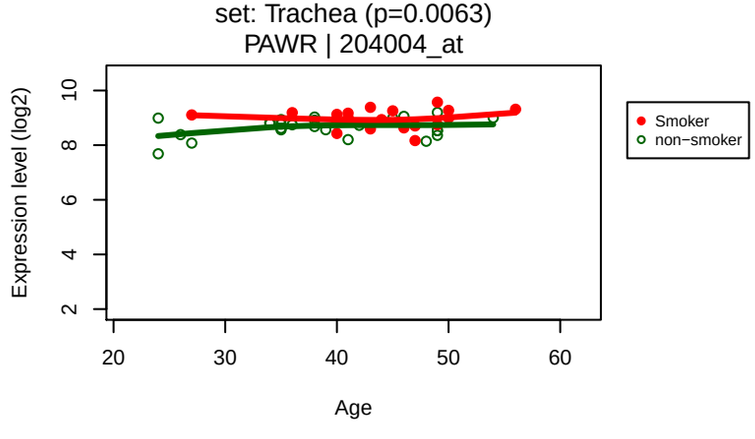

Gene expression ratio in Males
gene: PAWR | 204004_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1019
p=0.063
p=0.0336
p=0.0325
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAWR | 204004_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0725
p=0.3074
p=0.2829
p=0.5913
SmallAirway2
LargeAirway2
Trachea
Macrophage2

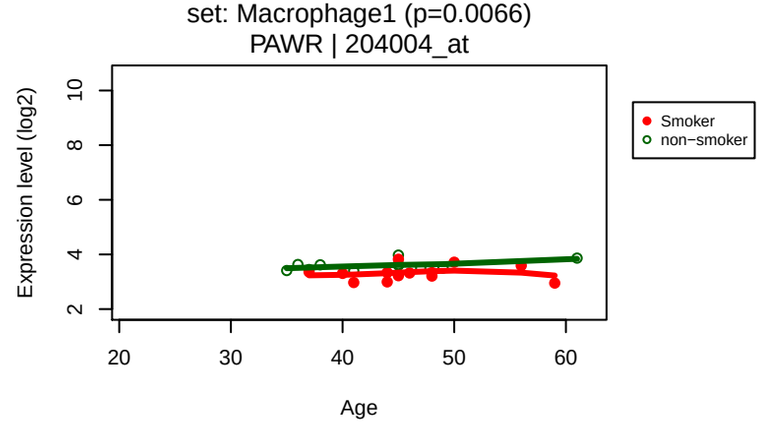

set: SmallAirway1 (p=0.1935)
PAWR | 204004_at
Expression level (log2)
Age
Smoker
non−smoker

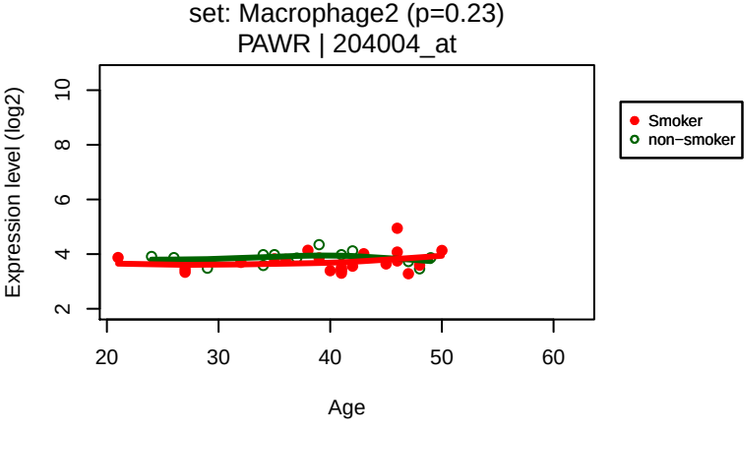

set: SmallAirway2 (p=0.0226)
PAWR | 204004_at
Expression level (log2)
Age
Smoker
non−smoker

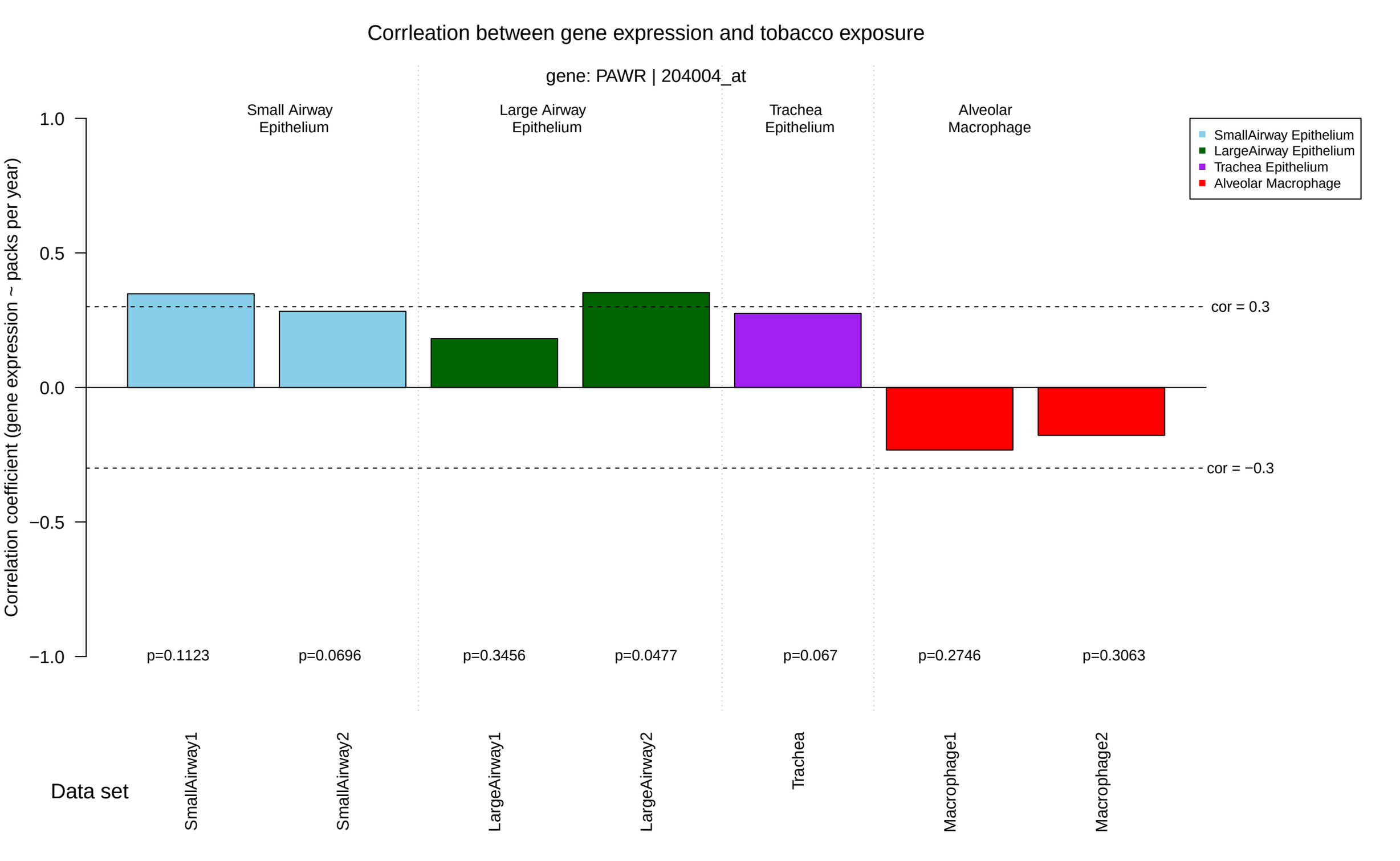

set: LargeAirway1 (p=0.739)
PAWR | 204004_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3345)
PAWR | 204004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0063)
PAWR | 204004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0066)
PAWR | 204004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.23)
PAWR | 204004_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAWR | 204004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1123
p=0.0696
p=0.3456
p=0.0477
p=0.067
p=0.2746
p=0.3063
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2